



Shared Computing Where PCI Performs and USB Lags

This article explores the evolution of shared computing, and evaluates the value proposition that a shared computing environment can offer to organisations. Also included are the test results conducted to gauge the efficacy of two methods (PCI and USB) that can be used for desktop virtualisation.

The genesis of shared computing dates back to the time when mainframes fed dumb clients. Then came the era of server-based computing, when these mainframes were replaced by servers and the earlier dumb clients by full-fledged PCs. Over time, thin clients have gained traction in many server-based computing set ups—where most, if not all applications are run on the server.

This server-based computing was intended to provide the same advantages as mainframe computing, while mitigating the cost and environmental factors—but it created a completely different set of disadvantages, the key one being the constrained user experience with limited desktop interface performance, especially when graphical applications are used. Apart from this, thin clients, which can be called scaled-down versions of PCs, invariably require special customisation and administration, expensive high-end server components, and so on.

Another option that has emerged to overcome these disadvantages, while still tapping the benefits of server-based computing, has been the zero client, based on the concept of desktop virtualisation. The concept behind these virtual desktops is to optimally use the power of the PC, which remains under-utilised in most cases, as most of us use a small fraction of its power. Desktop virtualisation enables a single

PC to be ‘virtualised’ (or shared) by many users—where all users get to experience their own individual computing session.

The relevance of desktop virtualisation for organisations

It is important to evaluate the suitability of desktop virtualisation for organisations of different sizes. There are a number of reasons and advantages that CTOs of companies of all sizes and markets can consider before adopting desktop virtualisation. Shared computing or desktop virtualisation is an ideal solution for companies moving to cloud and Web-based computing. It works fine for CIOs seeking to better manage PC proliferation and costs. It is ideal for education and public sector institutions seeking to provide desktop access to all constituencies at a low initial cost of purchase, and even for those who want to achieve sustainability through low on-going costs.

Other benefits of desktop virtualisation include ease of centralised management, enhanced security as a result of limited access to critical infrastructure, low cost of entry, and dramatically reduced life-cycle and maintenance costs. And, what if we use open source operating systems along with desktop virtualisation?